Exercise 1

a) Set $P(n) = \text{"}\sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6}\text{"}$

we prove by induction that $P(n)$ holds for every $n > 0$

Base Step $(n=1)$

$P(1) = \text{"}1^2 = \frac{1 \cdot 2 \cdot 3}{6}\text{"}$ which is true

Inductive Step $(P(n) \Rightarrow P(n+1))$

We assume $P(n)$ and prove $P(n+1)$.

$$\sum_{k=1}^{n+1} k^2 = \sum_{k=1}^{n} k^2 + (n+1)^2$$

$$= \frac{n(n+1)(2n+1)}{6} + (n+1)^2 \quad \text{By inductive hypotesis}$$

$$= \frac{2n^3 + 3n^2 + n}{6} + \frac{6n^2 + 12n + 6}{6}$$

$$= \frac{(n+1)(n+2)(2n+3)}{6} = \frac{(n+1)((n+1)+1)(2(n+1)+1)}{6},$$

as desired.

b) We claim that $a_n = 2^n \ \forall\, n \geq 0$

We set $P(n) = \text{"}a_n = 2^n\text{"}$ and prove that $P(n)$ holds for any $n \geq 0$ by strong induction

Base Step $(n=0)$

$P(0) = \text{"}a_0 = 1\text{"}$ which is true by definition of a_0

Inductive Step $(P(k) \ \forall\, 0 \leq k \leq n \Rightarrow P(n+1))$

$$a_{n+1} = a_n + a_{n-1} + \dots + a_0 + 1 \quad \text{By inductive hypotesis}$$

$$= 2^n + 2^{n-1} + \dots + 1 + 1 \quad \text{By the geometric sum formula seen in class}$$

$$= \frac{2^{n+1} - 1}{2 - 1} + 1$$

$$= 2^{n+1} \quad \text{as whished.}$$

Exercise 2

a) $$\binom{n}{k} + \binom{n}{k+1} = \frac{n \cdots (n-k+1)}{k \cdots 1} + \frac{n \cdots (n-k+1)(n-k)}{(k+1)k \cdots 1}$$

$$= \frac{n \cdots (n-k+1)}{k \cdots 1}\left(1 + \frac{n-k}{k+1}\right)$$

$$= \frac{(n+1)\, n \cdots (n-k+1)}{(k+1)\, k \cdots 1}$$

$$= \binom{n+1}{k+1}.$$

b) We set $P(n) = \text{"}(x+y)^n = \sum_{k=0}^{n} x^k y^{n-k} \binom{n}{k}\text{"}$ and we prove $P(n)$ for every $n \geq 0$ by induction

<u>Base Step</u> $(n=0)$

$P(0) = \text{"}(x+y)^0 = x^0 y^0 \binom{0}{0}\text{"}$

which is true because both terms are equal to 1.

<u>Inductive Step</u> $(P(n) \Rightarrow P(n+1))$

We know that $(x+y)^n = \sum_{k=0}^{n} x^k y^{n-k} \binom{n}{k}$

multiplying both sides by $(x+y)$ we get

$$(x+y)^{n+1} = x \sum_{k=0}^{n} x^k y^{n-k} \binom{n}{k} + y \sum_{k=0}^{n} x^k y^{n-k} \binom{n}{k}$$

$$= \sum_{k=0}^{n} x^{k+1} y^{n-k} \binom{n}{k} + \sum_{k=0}^{n} x^k y^{n-k+1} \binom{n}{k}$$

$$= x^{n+1}\binom{n}{n} + \sum_{k=0}^{n-1} x^{k+1} y^{n-k} \binom{n}{k} + y^{n+1}\binom{n}{0} + \sum_{k=1}^{n} x^k y^{n-k+1} \binom{n}{k}$$

change of name $k \mapsto k-1$

$$= x^{n+1}\binom{n+1}{0} + \sum_{k=1}^{n} x^k y^{n+1-k} \binom{n}{k-1} + x^k y^{n-k+1} \binom{n}{k} + x^{n+1}\binom{n+1}{n+1}$$

$$= y^{n+1}\binom{n+1}{0} + \sum_{k=1}^{n} x^k y^{n+1-k} \left[\binom{n}{k-1} + \binom{n}{k}\right] + x^{n+1}\binom{n+1}{n+1}$$

$$= y^{n+1}\binom{n+1}{0} + \sum_{k=1}^{n} x^k y^{n+1-k} \binom{n+1}{k} + x^{n+1}\binom{n+1}{n+1}$$

$$= \sum_{k=0}^{n+1} x^k y^{(n+1)-k} \binom{n+1}{k} \quad \text{as desired}$$

c) We just need to plug in b), $x = y = 1$ to get $(1+1)^n = \sum_{k=0}^{n} \binom{n}{k}$ and to plug in $x = -1$, $y = 1$ to get $(-1+1)^n = \sum_{k=0}^{n} (-1)^k \binom{n}{k}$.

Exercise 3

a) i) Since $a>0$ we have $a \geq 0$, and since $x \leq y$ we have $x \leq y$ then by the axioms we have

$ax \leq ay$

We only have to prove that $ax \neq ay$.

Suppose by contraddiction that $ax = ay$, since $a>0$, $a \neq 0$ and a^{-1} exists. We multiply $ax = ay$ by a^{-1}, getting $x=y$, contraddicting $x<y$.

Hence $ax < ay$.

ii) Since $0 \leq x'$ and $0 \leq x$ by the axioms we have

$0 \cdot x' = 0 \leq x \cdot x'$

Applying the axioms with $x=x$, $y=y$ and $a = x' \geq 0$

we get $x \cdot x' \leq y \cdot x'$

Applying a) i) with $x = x'$, $y = y'$ and $a = y > 0$

we get $x' \cdot y < y' \cdot y$

Chaining these inequalities we get

$0 \leq x \cdot x' \leq y \cdot x' < y' \cdot y$ as desired.

b) From the proposition seen in class we have that $x^2 \geq 0$ $\forall x \in K$ and $1 > 0$.

Summing the two (part i) of the proposition seen in class) in any any ordered field K.

$x^2 + 1 > 0$ $\forall x \in K$

In particular $x^2 + 1 \neq 0$ $\forall x \in K$.

The the algebraic equation (of second degree) $x^2 + 1 = 0$ has no solution in K, hence K is not algebraically closed

Exercise 4

a) Let S_A be the set of upper bounds for A, and S_B the one for B. Let's prove that $S_B \subseteq S_A$. In fact take $s \in S_B$, it means that $b \leq s$ $\forall b \in B$, since $A \subset B$ we have $a \leq s$ $\forall a \in A$ and it follows that $s \in S_A$.

Then $\min(S_B) \geq \min(S_A)$, i.e. $\sup(B) \geq \sup(A)$.

b) Let S' be the sets of lower bounds for $-A$. We prove that $S' = -S_A$. Take $s \in S_A$, then $s \geq a$ $\forall a \in A$, then $-s \leq -a$ $\forall a \in A$

hence $-s$ is a lower bound for $-A$ $\Rightarrow -s \leq S'$

Take $s' \in S'$, then $s' \leq -a$ $\forall a \in A \Rightarrow -s' \geq a$ $\forall a$

Exercise 4

a) Let S_A be the set of upper bounds for A, and S_B the one for B. We want to prove that $S_B \subseteq S_A$. Take $s \in S_B$, then $s \geq b \ \forall \ b \in B$, since $A \subset B$, $s \geq a \ \forall \ a \in A \Rightarrow s \in S_A$. Then $\min(S_B) \geq \min(S_A)$, i.e. $\sup(B) \geq \sup(A)$.

b) We prove that s is an upper bound for A, if and only if $-s$ is a lower bound for $-A$. This follows from the fact that $s \geq a \ \forall \ a \in A$, if and only if $-s \leq -a \ \forall \ a \in A$. At this point we just notice that the sets of lower bounds for $-A$ is $-S_A$. Then $\inf(-A) = \min(-S_A) = -\max(S_A) = -\sup(A)$.

Exercise 5

a) We first notice that $0 \in X$ and, by definition of X, $0 \leq x \ \forall x \in X$. Hence X has a minimum:
$0 = \min(X) = \inf(X)$ and X is bounded from below.

By definition of X, $x < 1 \ \forall x \in X$, hence X is bounded from above. At the same time for all $\varepsilon > 0$ ~~we have that $1 - \min\{\varepsilon/2, 1\} \in X$ and $1 - \min\{\varepsilon/2, 1\} > 1 - \varepsilon$, hence~~ we can find $x \in X$ s.t. $1 - \varepsilon < x$, hence $1 = \sup(X)$. Since $1 \notin X$, X has no maximum

b) $\frac{1}{n} \leq 1 \ \forall n > 0$, hence X is bounded from above by 1.
Moreover, for $n = 1$ $\frac{1}{n} = 1$, hence $\max(X) = \sup(X) = 1$.
$\forall n \geq 0$ $\frac{1}{n} > 0$ hence X is bounded from below by 1.
Moreover for any $\varepsilon > 0$ we can fin n big enough s.t. $\frac{1}{n} < \varepsilon$. Hence $\inf(X) = 0$. Since $0 \notin 0$, X has no minimum

c) We can immediately see that $\sin(\sqrt{n}) \leq 1$, equality holds only if $\sqrt{n} = \pi/2 + 2k\pi$, but this equation isn't satisfied for any n (the right hand side is not algebraic).
X is then bounded from above by 1.
Let's prove that $\sup(X) = 1$. From this it will follow that X has no maximum since $1 \notin X$.
Fix $\varepsilon > 0$. Notice that $\sqrt{n+1} - \sqrt{n} = \frac{1}{\sqrt{n+1} + \sqrt{n}} < \frac{1}{\sqrt{n}}$
For n big enough, let's say $n > n_0$, $\sqrt{n+1} - \sqrt{n} < \varepsilon$.
Let $\bar{n}$ be the smallest n s.t. $\sqrt{n} > 2\pi n_0 + \pi/2$
We can check that $\bar{n} > n_0$.
Since $\sqrt{\bar{n}} - \sqrt{\bar{n}-1} < \varepsilon$, we have $\left|\sqrt{\bar{n}} - (2\pi n_0 + \pi/2)\right| < \varepsilon$
Hence $\left|\sin(\sqrt{\bar{n}}) - \sin(2\pi n_0 + \pi/2)\right| = \left|\sin(\sqrt{\bar{n}}) - 1\right| < \varepsilon$.
Therefore $\sin(\sqrt{\bar{n}}) > 1 - \varepsilon \Rightarrow 1 = \sup(X)$.
With an equivalent argument one can prove that X is bounded from below, $\inf(X) = -1$ and X has no minimum.